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### Condensate collection assembly

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#### Abstract

The present disclosure relates to a heating, ventilation, and/or air conditioning (HVAC) system that includes a condensate collection assembly. The condensate collection assembly includes a drain pan configured to collect condensate generated by a heat exchanger of an HVAC system. The condensate collection assembly also includes a condensate capture receptacle configured to couple to the heat exchanger and overlap with the heat exchanger relative to a direction of air flow across the heat exchanger. The condensate capture receptacle is further configured to discharge condensate to the drain pan, and to be suspended above the drain pan relative to a direction of gravity.

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**Inventors:** Aly; Amr Maher (Wichita, KS), Cheng; Xianjin (Andover, KS), Romero; Elmer Wayne (Oxford, KS), Garrett; D'Marcus (Wichita, KS), Ergul; Selin Nur (Wichita, KS), Baljekar; Akshay (Pune, IN)

**Applicant:** Johnson Controls Tyco IP Holdings LLP (Milwaukee, WI)

**Family ID:** 1000008780250

**Assignee:** JOHNSON CONTROLS LIGHT COMMERCIAL IP GMBH (Neuhausen am Rheinfall, CH)

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## References Cited

### U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name | U.S. Cl. | CPC         |
|--------------|-------------|---------------|----------|-------------|
| 2020/0064054 | 12/2019     | Carlton       | N/A      | F28F 17/005 |
| 2020/0355439 | 12/2019     | Mercer        | N/A      | F28D 7/0066 |
| 2022/0349645 | 12/2021     | Romero et al. | N/A      | N/A         |

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*Primary Examiner:* Trpisovsky; Joseph F

*Attorney, Agent or Firm:* Fletcher Yoder, PC

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## Background/Summary

### BACKGROUND

(1) This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present disclosure, which are described below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it should be understood that these statements are to be read in this light, and not as admissions of prior art.

(2) Environmental control systems are utilized in residential, commercial, and industrial environments to control environmental properties, such as temperature and humidity, for occupants of the respective environments. The environmental control system may control the environmental properties through control of an air flow delivered to the environment. In some cases, environmental control systems include a vapor compression system, which includes heat exchangers, such as a condenser and an evaporator, that transfer thermal energy between the vapor compression system and the environment. Fans or blowers may direct a flow of supply air across a heat exchange area of the evaporator, and refrigerant circulating through the evaporator may absorb thermal energy from the supply air. Accordingly, the evaporator may discharge conditioned air, which is subsequently directed toward a cooling load, such as an interior of a building. In some instances, the evaporator may condense moisture suspended within the supply air, and condensate may form on an exterior surface of the evaporator. The condensate is generally directed to a drain pan configured to collect the condensate generated by the evaporator. However, the air flow passing across the evaporator may displace condensate formed and/or accumulated on the evaporator. Unfortunately, the velocity of the air flow may transfer the displaced condensate to a location or region beyond the drain pan.

### SUMMARY

(3) A summary of certain embodiments disclosed herein is set forth below. It should be understood

that these aspects are presented merely to provide the reader with a brief summary of these certain embodiments and that these aspects are not intended to limit the scope of this disclosure. Indeed, this disclosure may encompass a variety of aspects that may not be set forth below.

(4) The present disclosure relates to a condensate collection assembly. The condensate collection assembly includes a drain pan configured to collect condensate generated by a heat exchanger of a heating, ventilation, and air conditioning (HVAC) system. The condensate collection assembly also includes a condensate capture receptacle configured to couple to the heat exchanger. The condensate capture receptacle is further configured to overlap with the heat exchanger relative to a direction of air flow across the heat exchanger, to discharge condensate to the drain pan, and to be suspended above the drain pan relative to a direction of gravity.

(5) The present disclosure also relates to a condensate collection assembly comprising a condensate trough configured to couple to a downstream end of a heat exchanger disposed within an air flow path of a heating, ventilation, and air conditioning (HVAC) system, relative to a direction of air flow across the heat exchanger. The condensate trough is further configured to capture condensate generated by the heat exchanger and to be disposed within the air flow path and offset from a drain pan of the HVAC system.

(6) The present disclosure further relates to an HVAC system that includes a heat exchanger disposed within an air flow path and configured to condition an air flow directed across the heat exchanger. The HVAC system also includes a condensate collection assembly configured to capture condensate generated by the heat exchanger. The condensate collection assembly includes a condensate capture receptacle configured to couple to the heat exchanger and to overlap with the heat exchanger relative to a direction of air flow across the heat exchanger along the air flow path. The condensate collection assembly is further configured to discharge condensate to a drain pan disposed beneath the heat exchanger and to be suspended above the drain pan relative to a direction of gravity.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

(1) Various aspects of this disclosure may be better understood upon reading the following detailed description and upon reference to the drawings in which:

(2) FIG. 1 is a perspective view of an embodiment of a building incorporating a heating, ventilation, and/or air conditioning (HVAC) system in a commercial setting, in accordance with an aspect of the present disclosure;

(3) FIG. 2 is a perspective view of an embodiment of a packaged HVAC unit, in accordance with an aspect of the present disclosure;

(4) FIG. 3 is a perspective view of an embodiment of a split, residential HVAC system, in accordance with an aspect of the present disclosure;

(5) FIG. 4 is a schematic diagram of an embodiment of a vapor compression system used in an HVAC system, in accordance with an aspect of the present disclosure;

(6) FIG. 5 is a perspective view of an embodiment of an HVAC unit including a condensate collection assembly, in accordance with an aspect of the present disclosure;

(7) FIG. 6 is a partial perspective view of an embodiment of an HVAC unit including a condensate collection assembly, in accordance with an aspect of the present disclosure;

(8) FIG. 7 is a perspective view of an embodiment of a heat exchanger that may be disposed within the housing of an HVAC unit, in accordance with an aspect of the present disclosure;

(9) FIG. 8 is a perspective view of an embodiment of a condensate collection assembly, in accordance with an aspect of the present disclosure;

(10) FIG. 9 is a partial perspective view of an embodiment of an HVAC unit including a condensate

collection assembly, in accordance with an aspect of the present disclosure;

(11) FIG. 10 is a schematic diagram of an embodiment of a condensate collection assembly, illustrating capture of condensate via a condensate capture receptacle, in accordance with an aspect of the present disclosure;

(12) FIG. 11 is a perspective view of an embodiment of a condensate capture receptacle, in accordance with an aspect of the present disclosure;

(13) FIG. 12 is an exploded perspective view of an embodiment of a condensate capture receptacle, in accordance with an aspect of the present disclosure;

(14) FIG. 13 illustrates side views of embodiments of a condensate capture receptacle, in accordance with an aspect of the present disclosure; and

(15) FIG. 14 illustrates partial perspective views of embodiments of the condensate collection assembly, illustrating support brackets coupled to condensate capture receptacles, in accordance with an aspect of the present disclosure.

#### DETAILED DESCRIPTION

(16) One or more specific embodiments of the present disclosure will be described below. These described embodiments are examples of the presently disclosed techniques. Additionally, in an effort to provide a concise description of these embodiments, all features of an actual implementation may not be described in the specification. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be appreciated that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

(17) When introducing elements of various embodiments of the present disclosure, the articles “a,” “an,” and “the” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Additionally, it should be understood that references to “one embodiment” or “an embodiment” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features.

(18) As used herein, the terms “approximately,” “generally,” and “substantially,” and so forth, are intended to convey that the property value being described may be within a relatively small range of the property value, as those of ordinary skill would understand. For example, when a property value is described as being “approximately” equal to (or, for example, “substantially similar” to) a given value, this is intended to mean that the property value may be within  $\pm 5\%$ , within  $\pm 4\%$ , within  $\pm 3\%$ , within  $\pm 2\%$ , within  $\pm 1\%$ , or even closer, of the given value. Similarly, when a given feature is described as being “substantially parallel” to another feature, “generally perpendicular” to another feature, and so forth, this is intended to mean that the given feature is within  $\pm 5\%$ , within  $\pm 4\%$ , within  $\pm 3\%$ , within  $\pm 2\%$ , within  $\pm 1\%$ , or even closer, to having the described nature, such as being parallel to another feature, being perpendicular to another feature, and so forth. Further, it should be understood that mathematical terms, such as “planar,” “slope,” “perpendicular,” “parallel,” and so forth are intended to encompass features of surfaces or elements as understood to one of ordinary skill in the relevant art, and should not be rigidly interpreted as might be understood in the mathematical arts. For example, a “planar” surface is intended to encompass a surface that is machined, molded, or otherwise formed to be substantially flat or smooth (within related tolerances) using techniques and tools available to one of ordinary skill in the art. Similarly, a surface having a “slope” is intended to encompass a surface that is machined, molded, or otherwise formed to be oriented at an angle (e.g., incline) with respect to a point of reference using techniques and tools available to one of ordinary skill in the art.

(19) As briefly discussed above, a heating, ventilation, and/or air conditioning (HVAC) system may be used to thermally regulate a space within a building, home, or other suitable structure. For example, the HVAC system may include a vapor compression system that transfers thermal energy between a heat transfer fluid (e.g., a working fluid), such as a refrigerant, and a fluid to be conditioned, such as air. The vapor compression system includes heat exchangers (e.g. a condenser, an evaporator) that are fluidly coupled to one another via one or more conduits to form a refrigerant circuit. A compressor may be used to circulate the refrigerant through the refrigerant circuit and enable the transfer of thermal energy between components of the vapor compression system (e.g., the condenser, the evaporator) and one or more thermal loads (e.g., an environmental air flow, a supply air flow).

(20) Generally, one or more heat exchangers of the HVAC system may operate to condition a flow of air that is supplied to a conditioned space, such as interior of a building. The air to be conditioned may include ambient (e.g., outside) air, return air, a mixture of ambient air and return air, and/or another suitable flow of air. The HVAC system may include one or more fans or blowers that direct a flow of air across a heat exchange area of a heat exchanger to enable conditioning (e.g., heating, cooling, dehumidification) of the air. For example, an evaporator of the HVAC system may be configured to place refrigerant circulating through the evaporator in a heat exchange relationship with the air flow. In this way, the refrigerant within the evaporator may absorb thermal energy from the air flow, thereby cooling the air flow before the air flow is discharged toward a conditioned space as a supply air flow.

(21) Cooling of the air flow via the evaporator may cause moisture suspended within the air flow to condense, thereby forming condensate. In certain instances, condensate generated via the evaporator may initially collect on the heat exchange area of the evaporator. Condensate formed and/or accumulated on the evaporator may fall (e.g., via force of gravity) toward a drain pan positioned vertically beneath the evaporator. The drain pan may collect the condensate that falls from the evaporator and direct the condensate toward a drain or other suitable discharge outlet. Unfortunately, in some applications, the evaporator may be arranged (e.g., positioned, oriented) in a manner that reduces the effectiveness of conventional drain pans.

(22) For example, in some cases an HVAC system or HVAC unit having an evaporator may be arranged to direct an air flow across the evaporator in a generally lateral direction, such as either left to right or right to left. An air flow directed across the evaporator may cause condensate formed and/or accumulated thereon to become dislodged from the evaporator. More specifically, the air flow may dislodge the condensate from the evaporator and displace or carry the condensate in a direction (e.g., lateral direction) of the air flow. Moreover, increased turbulence and/or velocity of the air flow may cause the dislodged condensate to travel away from the evaporator (e.g., in a direction of the air flow). Condensate carried by the air flow may not fall, via force of gravity, into the drain pan positioned beneath the evaporator, which may result in the condensate permeating areas or regions external to the drain pan and/or HVAC unit. Therefore, it is now recognized that improved condensate collection systems are desirable for HVAC systems.

(23) Accordingly, embodiments of the present disclosure are directed to an improved condensate collection assembly that enables improved capture of condensate formed during operation of an HVAC system. For example, the condensate collection assembly may include one or more components configured to enable capture of condensate that may be susceptible to lateral discharge from an evaporator via an air flow directed across the evaporator in a lateral (e.g., horizontal, sideways) direction. As described in further detail below, the condensate collection assembly may include a drain pan and a condensate capture receptacle (e.g., a condensate shield, a condensate trough, a condensate covering) with one or more condensate ports (e.g., discharge ports, drainage ports). The condensate capture receptacle may be positioned at a downstream end of a heat exchanger (e.g., an evaporator) and may be configured to capture condensate generated during operation of the heat exchanger. To this end, the condensate collection assembly may include one

or more support brackets configured to couple the condensate capture receptacle to the heat exchanger. In particular, the one or more support brackets may support and suspend the condensate capture receptacle, such as above the drain pan relative to gravity. In this way, present embodiments enable capture of condensate dislodged from the heat exchanger, while also reducing or limiting air flow obstruction or interference caused by the condensate collection assembly.

(24) The condensate capture receptacle may be positioned on a lateral side (e.g., laterally adjacent to) the heat exchanger, such that the condensate capture receptacle may overlap with (e.g., shield) a lateral end of the heat exchanger relative to a direction of air flow across the heat exchanger.

Further, the condensate capture receptacle may be configured to block discharge of condensate in an outward (e.g., laterally outward) direction. In particular, the condensate capture receptacle may block discharge of condensate from a region or area generally defined by a perimeter of the drain pan. Additionally, the condensate capture receptacle may define a cavity configured to capture the discharged condensate blocked by the condensate capture receptacle. Condensate captured within the cavity may be directed toward the one or more condensate ports, and the one or more condensate ports may direct the condensate toward the drain pan for collection and discharge in a suitable manner. In this way, present embodiments enable improved capture and collection of condensate generated during operation of the HVAC system. Additional details and benefits enabled by the present embodiments are described in further detail below.

(25) Turning now to the drawings, FIG. 1 illustrates an embodiment of a heating, ventilation, and/or air conditioning (HVAC) system for environmental management that employs one or more HVAC units in accordance with the present disclosure. As used herein, an HVAC system includes any number of components configured to enable regulation of parameters related to climate characteristics, such as temperature, humidity, air flow, pressure, air quality, and so forth. For example, an “HVAC system” as used herein is defined as conventionally understood and as further described herein. Components or parts of an “HVAC system” may include, but are not limited to, all, some of, or individual parts such as a heat exchanger, a heater, an air flow control device, such as a fan, a sensor configured to detect a climate characteristic or operating parameter, a filter, a control device configured to regulate operation of an HVAC system component, a component configured to enable regulation of climate characteristics, or a combination thereof. An “HVAC system” is a system configured to provide such functions as heating, cooling, ventilation, dehumidification, pressurization, refrigeration, filtration, or any combination thereof. The embodiments described herein may be utilized in a variety of applications to control climate characteristics, such as residential, commercial, industrial, transportation, or other applications where climate control is desired.

(26) In the illustrated embodiment, a building **10** is air conditioned by a system that includes an HVAC unit **12** with a reheat system in accordance with present embodiments. The building **10** may be a commercial structure or a residential structure. As shown, the HVAC unit **12** is disposed on the roof of the building **10**; however, the HVAC unit **12** may be located in other equipment rooms or areas adjacent the building **10**. The HVAC unit **12** may be a single package unit containing other equipment, such as a blower, integrated air handler, and/or auxiliary heating unit. In other embodiments, the HVAC unit **12** may be part of a split HVAC system, such as the system shown in FIG. 3, which includes an outdoor HVAC unit **58** and an indoor HVAC unit **56**.

(27) The HVAC unit **12** is an air cooled device that implements a refrigeration cycle to provide conditioned air to the building **10**. Specifically, the HVAC unit **12** may include one or more heat exchangers across which an air flow is passed to condition the air flow before the air flow is supplied to the building. In the illustrated embodiment, the HVAC unit **12** is a rooftop unit (RTU) that conditions a supply air stream, such as environmental air and/or a return air flow from the building **10**. After the HVAC unit **12** conditions the air, the air is supplied to the building **10** via ductwork **14** extending throughout the building **10** from the HVAC unit **12**. For example, the ductwork **14** may extend to various individual floors or other sections of the building **10**. In certain

embodiments, the HVAC unit **12** may be a heat pump that provides both heating and cooling to the building with one refrigeration circuit configured to operate in different modes. In other embodiments, the HVAC unit **12** may include one or more refrigeration circuits for cooling an air stream and a furnace for heating the air stream.

(28) A control device **16**, one type of which may be a thermostat, may be used to designate the temperature of the conditioned air. The control device **16** also may be used to control the flow of air through the ductwork **14**. For example, the control device **16** may be used to regulate operation of one or more components of the HVAC unit **12** or other components, such as dampers and fans, within the building **10** that may control flow of air through and/or from the ductwork **14**. In some embodiments, other devices may be included in the system, such as pressure and/or temperature transducers or switches that sense the temperatures and pressures of the supply air, return air, and so forth. Moreover, the control device **16** may include computer systems that are integrated with or separate from other building control or monitoring systems, and even systems that are remote from the building **10**.

(29) FIG. **2** is a perspective view of an embodiment of the HVAC unit **12**. In the illustrated embodiment, the HVAC unit **12** is a single package unit that may include one or more independent refrigeration circuits and components that are tested, charged, wired, piped, and ready for installation. The HVAC unit **12** may provide a variety of heating and/or cooling functions, such as cooling only, heating only, cooling with electric heat, cooling with dehumidification, cooling with gas heat, or cooling with a heat pump. As described above, the HVAC unit **12** may directly cool and/or heat an air stream provided to the building **10** to condition a space in the building **10**.

(30) As shown in the illustrated embodiment of FIG. **2**, a cabinet **24** encloses the HVAC unit **12** and provides structural support and protection to the internal components from environmental and other contaminants. In some embodiments, the cabinet **24** may be constructed of galvanized steel and insulated with aluminum foil faced insulation. Rails **26** may be joined to the bottom perimeter of the cabinet **24** and provide a foundation for the HVAC unit **12**. In certain embodiments, the rails **26** may provide access for a forklift and/or overhead rigging to facilitate installation and/or removal of the HVAC unit **12**. In some embodiments, the rails **26** may fit into “curbs” on the roof to enable the HVAC unit **12** to provide air to the ductwork **14** from the bottom of the HVAC unit **12** while blocking elements such as rain from leaking into the building **10**.

(31) The HVAC unit **12** includes heat exchangers **28** and **30** in fluid communication with one or more refrigeration circuits. Tubes within the heat exchangers **28** and **30** may circulate refrigerant, such as R-**410A**, through the heat exchangers **28** and **30**. The tubes may be of various types, such as multichannel tubes, conventional copper or aluminum tubing, and so forth. Together, the heat exchangers **28** and **30** may implement a thermal cycle in which the refrigerant undergoes phase changes and/or temperature changes as it flows through the heat exchangers **28** and **30** to produce heated and/or cooled air. For example, the heat exchanger **28** may function as a condenser where heat is released from the refrigerant to ambient air, and the heat exchanger **30** may function as an evaporator where the refrigerant absorbs heat to cool an air stream. In other embodiments, the HVAC unit **12** may operate in a heat pump mode where the roles of the heat exchangers **28** and **30** may be reversed. That is, the heat exchanger **28** may function as an evaporator and the heat exchanger **30** may function as a condenser. In further embodiments, the HVAC unit **12** may include a furnace for heating the air stream that is supplied to the building **10**. While the illustrated embodiment of FIG. **2** shows the HVAC unit **12** having two of the heat exchangers **28** and **30**, in other embodiments, the HVAC unit **12** may include one heat exchanger or more than two heat exchangers.

(32) The heat exchanger **30** is located within a compartment **31** that separates the heat exchanger **30** from the heat exchanger **28**. Fans **32** draw air from the environment through the heat exchanger **28**. Air may be heated and/or cooled as the air flows through the heat exchanger **28** before being released back to the environment surrounding the HVAC unit **12**. A blower assembly **34**, powered

by a motor **36**, draws air through the heat exchanger **30** to heat or cool the air. The heated or cooled air may be directed to the building **10** by the ductwork **14**, which may be connected to the HVAC unit **12**. Before flowing through the heat exchanger **30**, the conditioned air flows through one or more filters **38** that may remove particulates and contaminants from the air. In certain embodiments, the filters **38** may be disposed on the air intake side of the heat exchanger **30** to prevent contaminants from contacting the heat exchanger **30**.

(33) The HVAC unit **12** also may include other equipment for implementing the thermal cycle. Compressors **42** increase the pressure and temperature of the refrigerant before the refrigerant enters the heat exchanger **28**. The compressors **42** may be any suitable type of compressors, such as scroll compressors, rotary compressors, screw compressors, or reciprocating compressors. In some embodiments, the compressors **42** may include a pair of hermetic direct drive compressors arranged in a dual stage configuration **44**. However, in other embodiments, any number of the compressors **42** may be provided to achieve various stages of heating and/or cooling. As may be appreciated, additional equipment and devices may be included in the HVAC unit **12**, such as a solid-core filter drier, a drain pan, a disconnect switch, an economizer, pressure switches, phase monitors, and humidity sensors, among other things.

(34) The HVAC unit **12** may receive power through a terminal block **46**. For example, a high voltage power source may be connected to the terminal block **46** to power the equipment. The operation of the HVAC unit **12** may be governed or regulated by a control board **48**. The control board **48** may include control circuitry connected to a thermostat, sensors, and alarms. One or more of these components may be referred to herein separately or collectively as the control device **16**. The control circuitry may be configured to control operation of the equipment, provide alarms, and monitor safety switches. Wiring **49** may connect the control board **48** and the terminal block **46** to the equipment of the HVAC unit **12**.

(35) FIG. 3 illustrates a residential heating and cooling system **50**, also in accordance with present techniques. The residential heating and cooling system **50** may provide heated and cooled air to a residential structure, as well as provide outside air for ventilation and provide improved indoor air quality (IAQ) through devices such as ultraviolet lights and air filters. In the illustrated embodiment, the residential heating and cooling system **50** is a split HVAC system. In general, a residence **52** conditioned by a split HVAC system may include refrigerant conduits **54** that operatively couple the indoor unit **56** to the outdoor unit **58**. The indoor unit **56** may be positioned in a utility room, an attic, a basement, and so forth. The outdoor unit **58** is typically situated adjacent to a side of residence **52** and is covered by a shroud to protect the system components and to prevent leaves and other debris or contaminants from entering the unit. The refrigerant conduits **54** transfer refrigerant between the indoor unit **56** and the outdoor unit **58**, typically transferring primarily liquid refrigerant in one direction and primarily vaporized refrigerant in an opposite direction.

(36) When the system shown in FIG. 3 is operating as an air conditioner, a heat exchanger **60** in the outdoor unit **58** serves as a condenser for re-condensing vaporized refrigerant flowing from the indoor unit **56** to the outdoor unit **58** via one of the refrigerant conduits **54**. In these applications, a heat exchanger **62** of the indoor unit functions as an evaporator. Specifically, the heat exchanger **62** receives liquid refrigerant, which may be expanded by an expansion device, and evaporates the refrigerant before returning it to the outdoor unit **58**.

(37) The outdoor unit **58** draws environmental air through the heat exchanger **60** using a fan **64** and expels the air above the outdoor unit **58**. When operating as an air conditioner, the air is heated by the heat exchanger **60** within the outdoor unit **58** and exits the unit at a temperature higher than it entered. The indoor unit **56** includes a blower or fan **66** that directs air through or across the indoor heat exchanger **62**, where the air is cooled when the system is operating in air conditioning mode. Thereafter, the air is passed through ductwork **68** that directs the air to the residence **52**. The overall system operates to maintain a desired temperature as set by a system controller. When the

temperature sensed inside the residence 52 is higher than the set point on the thermostat, or the set point plus a small amount, the residential heating and cooling system 50 may become operative to refrigerate additional air for circulation through the residence 52. When the temperature reaches the set point, or the set point minus a small amount, the residential heating and cooling system 50 may stop the refrigeration cycle temporarily. The outdoor unit 58 includes a reheat system in accordance with present embodiments.

(38) The residential heating and cooling system 50 may also operate as a heat pump. When operating as a heat pump, the roles of heat exchangers 60 and 62 are reversed. That is, the heat exchanger 60 of the outdoor unit 58 will serve as an evaporator to evaporate refrigerant and thereby cool air entering the outdoor unit 58 as the air passes over the outdoor heat exchanger 60. The indoor heat exchanger 62 will receive a stream of air blown over it and will heat the air by condensing the refrigerant.

(39) In some embodiments, the indoor unit 56 may include a furnace system 70. For example, the indoor unit 56 may include the furnace system 70 when the residential heating and cooling system 50 is not configured to operate as a heat pump. The furnace system 70 may include a burner assembly and heat exchanger, among other components, inside the indoor unit 56. Fuel is provided to the burner assembly of the furnace 70 where it is mixed with air and combusted to form combustion products. The combustion products may pass through tubes or piping in a heat exchanger, separate from heat exchanger 62, such that air directed by the blower 66 passes over the tubes or pipes and extracts heat from the combustion products. The heated air may then be routed from the furnace system 70 to the ductwork 68 for heating the residence 52.

(40) FIG. 4 is an embodiment of a vapor compression system 72 that can be used in any of the systems described above. The vapor compression system 72 may circulate a refrigerant through a circuit starting with a compressor 74. The circuit may also include a condenser 76, an expansion valve(s) or device(s) 78, and an evaporator 80. The vapor compression system 72 may further include a control panel 82 that has an analog to digital (A/D) converter 84, a microprocessor 86, a non-volatile memory 88, and/or an interface board 90. The control panel 82 and its components may function to regulate operation of the vapor compression system 72 based on feedback from an operator, from sensors of the vapor compression system 72 that detect operating conditions, and so forth.

(41) In some embodiments, the vapor compression system 72 may use one or more of a variable speed drive (VSDs) 92, a motor 94, the compressor 74, the condenser 76, the expansion valve or device 78, and/or the evaporator 80. The motor 94 may drive the compressor 74 and may be powered by the variable speed drive (VSD) 92. The VSD 92 receives alternating current (AC) power having a particular fixed line voltage and fixed line frequency from an AC power source, and provides power having a variable voltage and frequency to the motor 94. In other embodiments, the motor 94 may be powered directly from an AC or direct current (DC) power source. The motor 94 may include any type of electric motor that can be powered by a VSD or directly from an AC or DC power source, such as a switched reluctance motor, an induction motor, an electronically commutated permanent magnet motor, or another suitable motor.

(42) The compressor 74 compresses a refrigerant vapor and delivers the vapor to the condenser 76 through a discharge passage. In some embodiments, the compressor 74 may be a centrifugal compressor. The refrigerant vapor delivered by the compressor 74 to the condenser 76 may transfer heat to a fluid passing across the condenser 76, such as ambient or environmental air 96. The refrigerant vapor may condense to a refrigerant liquid in the condenser 76 as a result of thermal heat transfer with the environmental air 96. The liquid refrigerant from the condenser 76 may flow through the expansion device 78 to the evaporator 80.

(43) The liquid refrigerant delivered to the evaporator 80 may absorb heat from another air stream, such as a supply air stream 98 provided to the building 10 or the residence 52. For example, the supply air stream 98 may include ambient or environmental air, return air from a building, or a

combination of the two. The liquid refrigerant in the evaporator **80** may undergo a phase change from the liquid refrigerant to a refrigerant vapor. In this manner, the evaporator **80** may reduce the temperature of the supply air stream **98** via thermal heat transfer with the refrigerant. Thereafter, the vapor refrigerant exits the evaporator **80** and returns to the compressor **74** by a suction line to complete the cycle.

(44) In some embodiments, the vapor compression system **72** may further include a reheat coil. In the illustrated embodiment, the reheat coil is represented as part of the evaporator **80**. The reheat coil is positioned downstream of the evaporator heat exchanger relative to the supply air stream **98** and may reheat the supply air stream **98** when the supply air stream **98** is overcooled to remove humidity from the supply air stream **98** before the supply air stream **98** is directed to the building **10** or the residence **52**.

(45) It should be appreciated that any of the features described herein may be incorporated with the HVAC unit **12**, the residential heating and cooling system **50**, or other HVAC systems.

Additionally, while the features disclosed herein are described in the context of embodiments that directly heat and cool a supply air stream provided to a building or other load, embodiments of the present disclosure may be applicable to other HVAC systems as well. For example, the features described herein may be applied to mechanical cooling systems, free cooling systems, chiller systems, or other heat pump or refrigeration applications.

(46) As noted above, HVAC systems typically include a drain pan configured to collect condensate that may form during operation of the HVAC system. Unfortunately, conventional systems generally rely on the force of gravity to direct condensate from a heat exchanger to the drain pan, which may not adequately facilitate desirable collection of condensate. For example, HVAC systems oriented to direct air flow across a heat exchanger in a lateral direction (e.g., horizontal right configuration, horizontal left configuration) may be susceptible to peripheral or external discharge of condensate. In other words, conventional drain pans may not adequately capture condensate generated during operation of the HVAC system, and condensate may undesirably travel (e.g., via air flow directed across the heat exchanger) to regions or areas external to the drain pan. Accordingly, embodiments of the present disclosure are directed toward a condensate collection assembly that is configured to more efficiently capture condensate generated during HVAC system operation.

(47) With the foregoing in mind, FIG. **5** is a perspective view of an embodiment of an HVAC unit **100** that can be used in any suitable HVAC system, such as the HVAC unit **12** of FIG. **1**, the split, residential HVAC system **50** of FIG. **3**, and/or the vapor compression system **72** of FIG. **4**. Indeed, it should be noted that the HVAC unit **100** may include embodiments and/or components of the HVAC unit **12**, embodiments or components of the split, residential HVAC system **50**, a rooftop unit (RTU), an air handler or air handling unit, an outdoor unit, and indoor unit, or any other suitable HVAC system. To facilitate discussion, the HVAC unit **100** and its respective components may be described with reference to a longitudinal axis **102**, a vertical axis **104**, which is oriented relative to a direction of gravity, and a lateral axis **106**.

(48) In the illustrated embodiment, the HVAC unit **100** includes a housing **108** defining an air flow path **110** therethrough. A heat exchanger **112** is disposed within the housing **108** and along the air flow path **110**. The HVAC unit **100** is configured to direct an air flow **114** through the housing **108** and across the heat exchanger **112** to enable conditioning of the air flow **114**. For example, the heat exchanger **112** may be an evaporator or other cooling coil configured to cool the air flow **114** directed through the housing **108**. In the illustrated embodiment, the housing **108** and the heat exchanger **112** are arranged in a generally horizontal flow configuration. In other words, the housing **108** is configured to direct the air flow **114** in a generally horizontal direction (e.g., along the longitudinal axis **102**) along the air flow path **110** and across the heat exchanger **112**. In some embodiments, the air flow **114** may be directed in through the housing **108** and across the heat exchanger **112** in a first direction **116** (e.g., in a horizontal left configuration of the heat exchanger

**112)** or in a second direction **118** (e.g., in a horizontal right configuration of the heat exchanger **112**). In the illustrated embodiment, the HVAC unit **100** includes a blower **120** configured to draw the air flow **114** into the housing **108** and across the heat exchanger **112** and to discharge the air flow **114** from the housing **108**. The air flow **114** may enter the housing **108** as a pre-conditioned air flow, a return air flow, an ambient air flow, any combination thereof, or another suitable air flow. The air flow **114** discharged from the HVAC unit **100** may be directed toward a conditioned space, such as via ductwork fluidly coupled to the air flow path **110**.

(49) As mentioned above, the heat exchanger **112** is configured to condition the air flow **114** directed through the housing **108** and across the heat exchanger **112**. For example, the heat exchanger **112** may be an evaporator configured to cool the air flow **114**. In some instances, the housing **108** may also include an additional heat exchanger disposed within the air flow path **110**, such as a heating coil. In certain operating modes of the HVAC unit **100**, the heat exchanger **112** (e.g., evaporator, cooling coil) and additional heat exchanger (e.g., heating coil) may operate in conjunction with one another to dehumidify the air flow **114**. During cooling and/or dehumidification of the air flow **114**, moisture within the air flow **114** may condense to form condensate in the form of liquid droplets. For example, condensate may form and/or collect on the heat exchanger **112**.

(50) As will be appreciated, it is desirable to collect and discharge condensate generated during operation of the HVAC unit **100** in a suitable manner. For example, it is desirable to capture the condensate and drain the condensate along a suitable flow path, as well as mitigate transfer of condensate to unintended areas or region (e.g., within the housing **108**, external to the housing **108**). Accordingly, the HVAC unit **100** includes a condensate collection assembly **122** configured to capture and collect condensate generated from the air flow **114** during operation of the HVAC unit **100**. For example, the condensate collection assembly **122** is configured to capture and collect condensate that may form on the heat exchanger **112** and may fall from the heat exchanger **112** (e.g., along vertical axis **104**) due to force of gravity. The condensate collection assembly **122** is also configured to capture condensate that may be dislodged from the heat exchanger **112** (e.g., at least partially along longitudinal axis **102**) via the air flow **114** directed across the heat exchanger **112**. As described herein, condensate dislodged from the heat exchanger **112** by the air flow **114** and traveling at least partially in a direction of the air flow **114** may be referred to as “condensate blowoff” or “condensate carryover.” The condensate collection assembly **122** of the present disclosure is configured to capture condensate blowoff or carryover and block condensate exposure to unintended areas, such as certain areas within the housing **108**, the blower **120**, filters (e.g., filter **38**) within the housing **108**, ductwork (e.g., ductwork **14** and **68**) fluidly coupled to the housing **108**, and/or other unintended areas associated with the HVAC unit **100** (e.g., building **10**, residence **52**).

(51) In the illustrated embodiment, the condensate collection assembly **122** includes components disposed beneath (e.g., relative to vertical axis **104**) the heat exchanger **112**. The condensate collection assembly **122** also includes components disposed on or adjacent one or more lateral sides **124** (e.g., vertically-extending sides) of the heat exchanger **112**. In other words, the condensate collection assembly **122** includes components disposed laterally outward (e.g., relative to longitudinal axis **102** and/or lateral axis **106**) from the heat exchanger **112**. In the manner described below, the components of the condensate collection assembly **122** enable capture and collection of condensate that may fall from the heat exchanger **112** (e.g., along vertical axis **104**), as well as condensate that may be dislodged by the air flow **114** directed across the heat exchanger **112**. The condensate captured and collected by the condensate collection assembly **122** may then be discharged or drained from the HVAC unit **100** in a desirable manner.

(52) The embodiments discussed below with reference to FIGS. **6-10** are described in the context of the HVAC unit **100** arranged to direct the air flow **114** through the housing **108** of the HVAC unit **100** in the first direction **116**. In other words, the HVAC unit **100** and condensate collection

assembly **122** described as implemented with the HVAC unit **100** in a horizontal left configuration. However, it should be appreciated that embodiments of the condensate collection assembly **122** may be similarly implemented with the HVAC unit **100** oriented in a horizontal right configuration and with the air flow **114** directed through the housing **108** in the second direction **118** discussed above. Additionally, in some embodiments the condensate collection assembly **122** may be implemented with the HVAC unit **100** oriented in a vertical flow configuration (e.g., up flow configuration, down flow configuration), in which the HVAC unit **100** is oriented to direct the air flow **114** across the heat exchanger **112** generally along the vertical axis **104**.

(53) FIG. **6** is a perspective view of an embodiment of a portion of the HVAC unit **100**. For clarity and to better illustrate the condensate collection assembly **122**, the blower **120** and certain portions of the housing **108** are not shown. In some embodiments, the heat exchanger **112** disposed within the HVAC unit **100** may include one or more heat exchange slabs **202** (e.g., portions, segments, sections) disposed along the air flow path **110**. For example, the heat exchange slabs **202** may include tube and fin heat exchange coils, microchannel heat exchange tubes (e.g., finned microchannel heat exchange tubes), or another suitable type of heat exchanger. The heat exchanger **112** also includes one or more delta plates **204** coupled to the heat exchange slabs **202**. The one or more delta plates **204** may function to support the one or more heat exchange slabs **202** in a desired configuration (e.g., relative to one another). In the illustrated embodiment, the heat exchanger **112** includes two heat exchange slabs **202** arranged in an “A” coil configuration. However, in other embodiments, heat exchange slabs **202** may be arranged in a “V”, “N”, or “Z” configuration. In some embodiments, the heat exchanger **112** may include one heat exchange slab **202** disposed along the air flow path **110** and extending across the air flow path **110**. Indeed, the heat exchange slabs **202** may be oriented in any suitable configuration along the air flow path **110** relative to the lateral axis **106**, longitudinal axis **102**, and/or vertical axis **104**.

(54) In the illustrated A-coil configuration, the heat exchange slabs **202** (e.g., a first heat exchange portion **206** of the heat exchanger **112**, a second heat exchange portion **208** of the heat exchanger **112**) are disposed at an angle relative to one another and relative to the first direction **116** of the air flow **114** through the housing **108**. The first heat exchange portion **206** extends along the longitudinal axis **102** in the first direction **116** at a downward slope or angle relative to the vertical axis **104**, and the second heat exchange portion **208** extends along the longitudinal axis **102** in the first direction **116** at an upward slope or angle relative to the vertical axis **104**. Thus, the heat exchange slabs **202** converge towards one another in the first direction **116** along the longitudinal axis **102**, such that respective downstream ends **210** (e.g., relative to the first direction **116** of air flow **114**) of the first heat exchange portion **206** and the second heat exchange portion **208** define an apex **212** (e.g., a vertex of the A-coil configuration) of the heat exchanger **112**. During operation of the HVAC unit **100**, the air flow **114** may initially flow along the air flow path **110** between (e.g., relative to vertical axis **104**) respective upstream ends **214** (e.g., relative to the first direction **116** of air flow **114**) of the first heat exchange portion **206** and the second heat exchange portion **208**. As the air flow **114** travels along the air flow path **110**, the air flow **114** may pass across the heat exchange slabs **202**. Thereafter, the air flow **114** may continue traveling along the air flow path **110** through the housing **108** (e.g., toward the blower **120**).

(55) As mentioned above, each heat exchange slab **202** may include rows of tubes configured to direct refrigerant or working fluid therethrough. Each heat exchange slab **202** may also include fins coupled to and extending between the respective tubes. The fins may support or reinforce the arrangement of tubes and may also be configured to increase a heat transfer surface area of the heat exchange slabs **202**. During operation of the HVAC unit **100** to cool the air flow **114** via the heat exchanger **112**, heat may be transferred from the air flow **114** to the refrigerant circulated through the tubes of the heat exchange slabs **202**. As a result, moisture within the air flow **114** may condense to form condensate (e.g., water droplets) that may collect on heat exchange slabs **202** (e.g., tubes, fins, and/or the delta plates **204**). In some instances, the velocity and/or turbulence of

the air flow **114** may cause condensate formed on one or more of the heat exchange slabs **202** to travel at least partially in the first direction **116**. That is, the air flow **114** may drive or force condensate formed on the heat exchange slabs **202** to become dislodged from the heat exchange slabs **202** and migrate in the first direction **116** (e.g., along the fins and/or tubes, toward the apex **212** of the heat exchanger **112**).

(56) To capture and collect condensate generated via operation of the heat exchanger **112**, the HVAC unit **100** includes the condensate collection assembly **122**. As shown, the condensate collection assembly **122** may be configured to couple to one or more of the heat exchange slabs **202**, one or more of the delta plates **204**, or both. In the illustrated embodiment, the condensate collection assembly **122** includes a drain pan **220** disposed within the housing **108** and beneath the heat exchanger **112** relative to the vertical axis **104**. The condensate collection assembly **122** also includes a condensate capture receptacle **222** disposed within the housing **108** and adjacent one or more lateral sides **124** of the heat exchanger **112**. Additionally, the condensate collection assembly **122** may include one or more support brackets **226** configured to couple the condensate capture receptacle **222** to one or more of the heat exchange slabs **202**, one or more of the delta plates **204**, or both. That is, the support brackets **226** may support and suspend the condensate capture receptacle **222** from the drain pan **220**. In this way, the condensate capture receptacle **222** may be suspended above the drain pan **220** relative to the vertical axis **104**, and the condensate capture receptacle **222** may be disposed within the air flow path **110**.

(57) As illustrated in FIG. 6, the condensate capture receptacle **222** may extend from a first support bracket **230** to a second support bracket **232** and across the air flow path **110** extending through the housing **108**. Furthermore, the condensate capture receptacle **222** may be disposed at least partially laterally outward from the heat exchanger **112** (e.g., along longitudinal axis **102** and/or lateral axis **106**). In addition, the condensate capture receptacle **222** may be at least partially disposed within (e.g., across) the air flow path **110** defined by the housing **108**. The condensate capture receptacle **222** is coupled to the heat exchanger **112** (e.g., via the support brackets **226**) and is disposed within an outer perimeter **224** defined by the drain pan **220**. Accordingly, the condensate capture receptacle **222** may be configured to direct captured condensate into the drain pan **220**. Details of the condensate collection assembly **122** are described further below with respect to FIG. 8.

(58) In addition to efficiently capturing condensate generated during operation of the HVAC unit **100**, the condensate capture receptacle **222** may also improve overall heat exchange efficiency of the HVAC unit **100** by blocking portions of the air flow **114** from exiting the heat exchanger **112** through the apex **212** of the heat exchanger **112** and diverting (e.g., redirecting) the portions of the air flow **114**, such that the portions of the air flow **114** flow across other portions of the first and/or second heat exchange slabs **202** instead of through the apex **212** of the heat exchanger **112**. In particular, as described above, each of the heat exchange slabs **202** may include heat exchange tubes configured to direct refrigerant therethrough, and the heat exchange slabs **202** may include fins extending between the heat exchange tubes. The fins extending from and between the heat exchange tubes may be configured to increase a heat transfer surface area of the heat exchange slabs **202**, and thus increase the amount of total heat transfer of the heat exchanger **112**.

(59) However, the apex **212** of the heat exchanger **112**, or any point at which two or more heat exchange slabs **202** converge (e.g., such as convergent points of the “V”, “N”, or “Z” configurations), may include a gap without heat exchange tubes between respective converging heat exchange slabs **202**. For example, two separate heat exchange slabs **202** may converge at the apex **212** and may define a gap or space of the heat exchanger **112** at the apex **212**. Alternatively, some embodiments of the heat exchanger **112** may include a section of heat exchange tubes that do not include fins. For example, the heat exchanger **112** may include microchannel tubes that form multiple heat exchange slabs **202**. That is, microchannel tubes may extend along both the first heat exchange portion **206** and the second heat exchange portion **208** to define multiple heat exchange slabs **202** that are fluidly coupled to one another at the apex **212**. To this end, the microchannel

tubes extending from the first heat exchange portion **206** to the second heat exchange portion **208** may be bent or curved at the apex **212**. In such an embodiment, the heat exchanger **112** may not include fins extending between the heat exchange tubes (e.g., microchannel tubes) at the apex **212** to facilitate bending of the heat exchange tubes extending from one heat exchange slab **202** to another heat exchange slab **202**. As a result, portions of the air flow **114** that travel through the apex **212** of the heat exchanger **112** may experience less efficient heat transfer as compared to portions of the air flow **114** that flow across the heat exchange slabs **202** at locations having heat exchange tubes with fins. As illustrated in FIG. **6**, the condensate capture receptacle **222** may be positioned (e.g., installed, mounted, coupled, directly coupled) to the heat exchanger **112** such that the condensate capture receptacle **222** covers (e.g., encases, shields, overlaps) the apex **212** of the heat exchanger **112** (e.g., relative to the air flow **114** directed in the first direction **116**). In particular, the condensate capture receptacle **222** may block the air flow **114** from crossing the heat exchanger **112** at the apex **212**, which may be a region of the heat exchanger **112** without heat exchange tubes (e.g., a gap between the converging heat exchange slabs **202**) or may be a region of the heat exchanger **112** with heat exchange tubes (e.g., microchannel tubes) that do not include fins. (60) By covering or overlapping with the apex **212** of the heat exchanger **112**, the condensate capture receptacle **222** may block portions of the air flow **114** from exiting the heat exchanger **112** through the apex **212** and may instead divert (e.g., redirect) those portions of the air flow **114** to exit the heat exchanger **112** by flowing across the first and/or second heat exchange slabs **202** instead of through the apex **212**. Therefore, including the condensate capture receptacle **222** to the heat exchanger **112** may increase the overall heat transfer efficiency of the heat exchanger **112** by causing a greater percentage of the air flow **114** to travel across (e.g., interact with) portions of heat exchange tubes having fins as compared to an HVAC unit operating without the condensate capture receptacle **222**. Thus, the condensate capture receptacle **222** may improve the performance of the heat exchanger **112** during operation in addition to efficiently and effectively capturing condensate generated by the heat exchanger **112**.

(61) FIG. **7** is a perspective view of an embodiment of the heat exchanger **112** that may be disposed within the housing **108** of the HVAC unit **100**. As noted above, FIG. **7** is described below in the context of the HVAC unit **100** arranged to direct the air flow **114** through the housing **108** of the HVAC unit **100** in the first direction **116**. The heat exchanger **112** of the illustrated embodiment is a microchannel heat exchanger **300** having a plurality of microchannel tubes **308** defining an “A” configuration, as similarly described above. That is, the microchannel heat exchanger **300** includes a first section **306** (e.g., first heat exchange slab **202**) and a second section **312** (e.g., second heat exchange slab **202**) arranged to define the “A” configuration. Each microchannel tube **308** of the plurality of microchannel tubes **308** is included in both the first section **306** and the second section **312**. For example, each microchannel tube **308** may include a first portion **310** included in the first section **306** of the microchannel heat exchanger **300** and a section portion **314** included in the second section **312** of the microchannel heat exchanger **300**. The first portion **310** and the second portion **314** of each microchannel tube **308** may be fluidly coupled to one another via a bent portion **302** of the respective microchannel tube **308**. Thus, the first portion **310** and the second portion **314** of the microchannel heat exchanger **300** are fluidly coupled to one another.

(62) The bent portions **302** of the plurality of microchannel tubes **308** may define an apex **304** of the microchannel heat exchanger **300**. To facilitate manufacturing of the microchannel heat exchanger **300**, the bent portions **302** of the microchannel tubes **308** may not include fins extending therebetween. Specifically, the microchannel tubes **308** may be bent at the bent portions **302** and may be angled, twisted, or otherwise manipulated to enable desired packaging or arrangement of the microchannel tubes **308** relative to one another in the “A” configuration without compromising (e.g., blocking, restricting) internal flow paths (e.g., microchannels) of the microchannel tubes **308**. As a result, the microchannel tubes **308** may not include fins extending therebetween at the bent portions **302** (e.g., the apex **304**). However, the first section **306** of the microchannel heat

exchanger **300** may include fins extending between the first portions **310** of the microchannel tubes **308**, and the second section **312** of the microchannel heat exchanger **300** may include fins extending between the second portions **314** of the microchannel tubes **308**.

(63) As similarly discussed above, the first and second sections **306**, **312** of the microchannel heat exchanger **300** may be disposed at an angle relative to one another and relative to the first direction **116** of the air flow **114** directed through the housing **108** and across the microchannel heat exchanger **300**. The first portions **310** of the microchannel tubes **308** in the first section **306** may extend along the longitudinal axis **102** at a downward slope or angle relative to the vertical axis **104**, and the second portions **314** of the microchannel tubes **308** in the second section **312** may extend along the longitudinal axis **102** at an upward slope or angle relative to the vertical axis **104**. Thus, the first and second sections **306**, **312** of the microchannel heat exchanger **300** converge towards one another in the first direction **116** along the longitudinal axis **102**, such that respective downstream ends **316** (e.g., relative to the first direction **116** of air flow **114**) of the first and second sections **306**, **312** of the microchannel heat exchanger **300** may partially define the apex **304** (e.g., a vertex of the “A” configuration) of the microchannel heat exchanger **300**.

(64) As mentioned above, each microchannel tube **308** may include a plurality of channels or flow paths (e.g., internal flow paths, microchannels) formed therethrough to direct a flow of refrigerant through the microchannel tubes **308** (e.g., through the first portion **310**, the bent portion **302**, and the second portion **314**). During operation, the refrigerant may flow into the first portions **310** of the microchannel tubes **308**, and may be directed generally in a first direction of flow **320** through the first section **306** of the microchannel heat exchanger **300** toward the apex **304** of the microchannel heat exchanger **300**. The refrigerant may then flow generally in a second direction of flow **322** through the bent segments **302** of the microchannel tubes **308** that curves around the apex **304** of the microchannel heat exchanger **300**. Thereafter, the refrigerant may flow into the second portions **314** of the microchannel tubes **308**, and may be directed generally in a third direction of flow **324** through the second section **312** of the microchannel heat exchanger **300**. Thereafter, the refrigerant may be discharged from the microchannel heat exchanger **300**. It should be appreciated that, in some embodiments, the refrigerant may be directed to flow through the microchannel tubes **308** in an direction opposite that described above. For example, the refrigerant may flow into the second portions **314** of the microchannel tubes **308**, flow along the second section **312** toward the bent portions **302** of the microchannel tubes **308**, flow around the apex **304** of the microchannel heat exchanger **300**, and then flow through the first portions **310** of the microchannel tubes **308** and along the first section **306** before discharged from the microchannel heat exchanger **300**.

(65) In the illustrated embodiment, each microchannel tube **308** may have a generally ribbon shape (e.g., a width of the microchannel tube **308** is greater than a thickness or height of the microchannel tube **308**), and each microchannel tube **308** may be positioned within the first and second sections **306**, **312** of the microchannel heat exchanger **300** such that a width of each microchannel tube **308** extends along the vertical axis **104**. Additionally, each microchannel tube **308** may extend continuously from an upstream end **326** of the first section **306** of the microchannel heat exchanger **300**, around the apex **304**, and to an upstream end **326** of the second section **312** of the microchannel heat exchanger **300**. To this end, a portion of each microchannel tube **308** at the bent portion **302** that is bent around the apex **304** may be additionally rotated (e.g., twisted) so as to prevent crimping (e.g., closing) one or more of the fluid channels within the bent portions **302** of the microchannel tubes **308** at the apex **304**. Furthermore, as mentioned above, each the first and second sections **306**, **312** of the microchannel heat exchanger **300** may include a respective sets of fins **328** extending from and between the microchannel tubes **308** to increase heat transfer efficiency of the microchannel heat exchanger **300**. The fins **328**, **330** may additionally support and/or provide structural reinforcement to the microchannel tubes **308**. However, the bent portions **302** of the microchannel tubes **308** may be fin-less or bare so as to facilitate the rotation and bending of the bent portions **302** of the microchannel tubes **308** around the apex **304** of the

microchannel heat exchanger **300**.

(66) Furthermore, during operation of the HVAC unit **100**, and with additional reference to FIG. **6**, the air flow **114** may initially flow along the air flow path **110** between (e.g., relative to vertical axis **104**) the respective upstream ends **326** (e.g., relative to the first direction **116** of air flow **114**) of the first and second sections **306**, **312** of the microchannel heat exchanger **300**. As the air flow **114** travels along the air flow path **110**, the air flow **114** may pass across the first and second portions **310**, **314** of the microchannel tubes **308**. However a portion of the air flow **114** may pass across the apex **304** and thus across the bent portions **302** (e.g., fin-less portions) of the microchannel tubes **308** that curve around the apex **304**. As discussed above, fin-less portions of heat exchange tubes (e.g., the bent portions **302** of the microchannel tubes **308**) of the heat exchanger **112** may provide relatively less heat transfer compared to finned portions of heat exchange tubes (e.g., the first and second portions **310**, **314** of the microchannel tubes **308**) of the heat exchanger **112**. Therefore, including the condensate capture receptacle **222** with the microchannel heat exchanger **300** of FIG. **7** may increase the overall heat transfer efficiency of the microchannel heat exchanger **300** by causing a greater percentage of the air flow **114** to travel across (e.g., interact with) the first and second portions **310**, **314** of the microchannel tubes **308** having fins. Thus, the condensate capture receptacle **222** may improve the performance of the microchannel heat exchanger **300** during operation in addition to efficiently and effectively capturing condensate generated by the microchannel heat exchanger **300**, as discussed herein.

(67) Due to the additional benefits of improved (e.g., increased) overall heat transfer of the heat exchanger **112** (e.g., microchannel heat exchanger **300**), it should be appreciated that, in some embodiments, the condensate collection assembly **122** (e.g., the condensate capture receptacle **222**) may be included in HVAC units **100** oriented in any suitable direction. For example, in some embodiments, the heat exchanger **112** (e.g., microchannel heat exchanger **300**) may be oriented in a vertical configuration, in which the heat exchanger **112** may be positioned such that the apex **212** (e.g., apex **304**) of the heat exchanger **112** is facing up and is positioned at a top of the heat exchanger **112** with respect to the vertical axis **104**. Alternatively, in other embodiments, the apex **212** of the heat exchanger **112** may be facing down and is positioned at a bottom of the heat exchanger **112** with respect to the vertical axis **104**. It should be appreciated that the condensate collection assembly **122** (e.g., the condensate capture receptacle **222**) may increase overall heat transfer, and thus improve the performance of the heat exchanger **112** oriented in any direction.

(68) FIG. **8** is a perspective view of an embodiment of the condensate collection assembly **122**, which may be disposed within the HVAC unit **100** or any other suitable HVAC system having the heat exchanger **112** that may generate condensate. As mentioned above, the condensate collection assembly **122** may include the drain pan **220**, the condensate capture receptacle **222**, and the support brackets **226**. In the illustrated embodiment, the drain pan **220** includes a body portion **400** (e.g., a base, base surface, condensate collection surface) defining and extending along a length **402** of the drain pan **220** from a first end portion **406** (e.g., longitudinal end, longitudinal side) to a second end portion **408** (e.g., longitudinal end, longitudinal side) of the drain pan **220**. For clarity, it should be noted that the length **402** may extend generally along to the longitudinal axis **102**. The body portion **400** also defines and extends along a width **404** of the drain pan **220** from a third end portion **410** (e.g., lateral end, lateral side) to a fourth end portion **412** (e.g., lateral end, lateral side), and the width **404** may extend generally along to the lateral axis **106**.

(69) The drain pan **220** also includes a plurality of walls **414** (e.g., side walls) extending from the body portion **400**. In particular, the drain pan **220** includes a first wall **416** extending from the body portion **400** at the first end portion **406**, a second wall **418** extending from the body portion **400** at the second end portion **408**, a third wall **420** extending from the body portion **400** at the third end portion **410**, and a fourth wall **422** extending from the body portion **400** at the fourth end portion **412**. The first, second, third, and fourth walls **416**, **418**, **420**, and **422** may generally define the outer perimeter **224** of the drain pan **220**. The body portion **400** and the plurality of walls **414**

cooperatively define a basin **424** (e.g., reservoir, cavity, container, receptacle) of the drain pan **220** that is configured to capture and collect condensate generated during operation of the heat exchanger **112**. For example, condensate may fall from the heat exchanger **112** (e.g., along vertical axis **104**, via gravity) into the basin **424**. Additionally, in the manner described in further detail below, the condensate capture receptacle **222** may capture condensate (e.g., dislodged from the heat exchanger **112** via the air flow **114**) and direct the condensate into the basin **424**.

(70) The body portion **400** of the drain pan **220** defines or includes a draining surface **430**. The first, second, third, and fourth walls **416**, **418**, **420**, and **422** extend from the draining surface **430** and are configured to retain condensate directed into the basin **424**. The drain pan **220** further includes a drain port **426** (e.g., condensate port, discharge port) configured to enable discharge of condensate collected within the basin **424**. For example, the drain port **426** may be fluidly coupled to a discharge conduit configured to direct condensate toward a location external to the HVAC unit **100**. In the illustrated embodiment, the drain port **426** is formed in and/or coupled to the fourth wall **420** of the drain pan **220**, but in other embodiments the drain port **426** may be disposed in another suitable location of the drain pan **220**.

(71) In some embodiments, the draining surface **430** of drain pan **220** may include or define a slope (e.g., a compound slope) that enables drainage of condensate collected within the basin **424**. In other words, one or more portions of the draining surface **430** may be disposed at an angle relative to horizontal (e.g., sloped downwardly relative to gravity) to promote flow of condensate along the draining surface **430** generally towards the drain port **426**. In this way, condensate collected within the drain pan **220** may be suitably discharged from the HVAC unit **100**. For example, the draining surface **430** may be sloped downwardly (e.g., with respect to gravity, relative to vertical axis **104**) toward the drain port **426**, such that force of gravity may direct condensate accumulated on the draining surface **430** toward the drain port **426**. In some embodiments, one or more portions of the draining surface **430** may include a compound slope angled downwardly along the length **402** of the drain pan **220**. In some embodiments, the draining surface **430** may be sloped (e.g., downwardly sloped relative to gravity and/or vertical axis **104**) along the width **404** of the drain pan **220**, such as from the third wall **420** to the fourth wall **422** (e.g., along lateral axis **106**). Accordingly, the compound slope of the draining surface **430** may enable condensate collected on the draining surface **430** to flow generally along a direction of decline of the draining surface **430** toward the drain port **426**. In such embodiments, the drain port **426** may be located at a lower-most portion (e.g., relative to gravity and/or vertical axis **104**) of the draining surface **430**. In this manner, the drain pan **220** may be configured to promote drainage and/or discharge of condensate from the basin **424** via the drain port **426**.

(72) As illustrated in FIG. 8, the condensate collection assembly **122** also includes the condensate capture receptacle **222**. In some embodiments, the condensate capture receptacle **222** may be positioned adjacent one or more lateral sides **124** of the heat exchanger **112**. For example, the condensate capture receptacle **222** may be positioned at an end of the heat exchanger **112**. Referring back to FIG. 6, the condensate capture receptacle **222** is illustrated as disposed at a downstream end of the heat exchanger **112** relative to air flow **114** in the first direction **116** (e.g., a horizontal left configuration of the HVAC unit **100**). However, it should be noted that the installed configuration of the condensate collection assembly **122** shown in FIG. 6 may be similarly utilized with embodiments of the HVAC unit **100** oriented in a horizontal right configuration (e.g., to direct the air flow **114** in the second direction **118**), a vertical up flow configuration, or a vertical down flow configuration.

(73) Further, the condensate capture receptacle **222** may be configured to couple to the heat exchanger **112** (e.g., with the heat exchanger **112** at least partially supporting a weight of the condensate capture receptacle **222**). To this end, the condensate capture receptacle **222** may be configured to couple (e.g., mount) to one or more components of the heat exchanger **112**. The condensate capture receptacle **222** is also configured to capture condensate that may be generated

via operation of the heat exchanger **112**. In particular, the condensate capture receptacle **222** may be configured to capture condensate that forms on heat exchange coils, fins, and/or tubes and is subsequently dislodged from the heat exchanger **112** via the air flow **114** (e.g., traveling in the first direction **116**). Condensate captured by the condensate capture receptacle **222** may then be directed (e.g., via features of the condensate capture receptacle **222** and/or via gravity) into the basin **424** of the drain pan **220** positioned beneath the heat exchanger **112**.

(74) As illustrated in FIG. **8**, the condensate collection assembly **122** also includes the support brackets **226**. As discussed above, the support brackets **226** may couple (e.g., directly couple) the condensate capture receptacle **222** to the heat exchanger **112**, such as to the downstream end of heat exchanger **112** (e.g., with respect to the first direction **116** of the air flow **114**). In particular, a first portion **440** of each support bracket **226** may couple (e.g., directly couple) to the condensate capture receptacle **222**, and a second portion **442** of each support bracket **226** may couple (e.g., directly couple) to the heat exchanger **112** or to a component of the heat exchanger **112** (e.g., one or more of the heat exchange slabs **202**, one or more of the delta plates **204**, or both). The support brackets **226** may enable positioning of the condensate capture receptacle **222** to overlap with the apex **304** of the heat exchanger **112**, and thus overlap with (e.g., cover, enclose) the bent portions **302** of heat exchange tubes at the apex **304** relative to the first direction **116**. Furthermore, the support brackets **226** may enable adjustable positioning of the condensate capture receptacle **222** relative to the heat exchanger **112**. In this way, the support brackets **226** may enable use of the condensate capture receptacle **222** with heat exchangers **112** of varying size and/or capacity. For example, the support brackets **226** (e.g., the first portion **440** and/or the second portion **442**) may have one or more elongated slots **444** formed therethrough to enable adjustability in mounting the condensate capture receptacle **222** to the heat exchanger **112**. The first and/or second portions **440**, **442** of the support bracket **226** may be mounted (e.g., secured, coupled) respectively to the condensate capture receptacle **222** and/or the heat exchanger **112** via one or more fasteners. In addition, it should be understood that the support brackets **226** may be formed from any suitable material so as to enable secure attachment of the condensate capture receptacle **222** to the heat exchanger **112** (e.g., sheet metal, plastic).

(75) Continuing with reference to FIG. **8**, it should be understood that the support brackets **226** may be formed in any suitable shape so as to enable the condensate capture receptacle **222** to be mounted (e.g., secured) to the heat exchanger **112**. For example, as illustrated in FIG. **8**, a first support bracket **230** of the support brackets **226** is illustrated with the first portion **440** extending in a first plane and the second portion **442** extending in a second plane, different than the first plane. The first and second portions **440**, **442** are coupled via a third portion that extends between and cross-wise to the first and second portions **440**, **442**. However, a second support bracket **232** of the support brackets **226** is illustrated as including respective first and second portions **440**, **442** both extending in a single or common plane.

(76) The configuration of the support brackets **226** illustrated in FIG. **8** enable the condensate capture receptacle **222** to be secured to a microchannel heat exchanger, such as the microchannel heat exchanger **300** illustrated in FIG. **7**. Due to the rotating and bending of the microchannel tubes **308** at the apex **304** of the microchannel heat exchanger **300**, as discussed above, the bent portions **302** of the microchannel tubes **308** may flare (e.g., extends) outward (e.g., along lateral axis **106**) by an additional width on at least one lateral side of the microchannel heat exchanger **300**. In other words, the apex **304** of the microchannel heat exchanger **300** (e.g., formed by the bent portions **302** of the microchannel tubes **308**) may have a greater width than a width of the microchannel heat exchanger **300** measured across the first and second sections **306**, **312** of the microchannel heat exchanger **300**. Thus, the shape of the first and second support brackets **230**, **232** may enable positioning of the condensate capture receptacle **222** to overlap with an entire width of the apex **304** of the microchannel heat exchanger **300** while also coupling (e.g., directly coupling) the condensate capture receptacle **222** to the lateral downstream ends **316** of the microchannel heat

exchanger **300**. In other words, a first lateral side **450** of the condensate capture receptacle **222** may be mounted to (e.g., flush with) a lateral side of the heat exchanger **112** (e.g., via the second support bracket **232**), while a second lateral side **452** of the condensate capture receptacle **222** may be mounted to (e.g., not flush with) another lateral side of the heat exchanger **112** (e.g., via the first support bracket **230**). Details of the condensate capture receptacle **222** are described further below.

(77) Continuing with reference to FIG. **8**, the condensate capture receptacle **222** includes a shield panel **460** (e.g., condensate barrier, capture panel, blocking plate, impingement surface, shield surface), a first side panel **462** (e.g., side wall, at the first lateral side **450**), a second side panel **465** (e.g., side wall, on the second lateral side **452**), a head panel **464** (e.g., upper wall), a base panel **466** (e.g., lower wall), opposite the head panel **464** with respect to the vertical axis **104**, one or more condensate ports **470** (e.g., drainage ports) fluidly coupled to the base panel **466** and formed therethrough, and one or more flanges **468** (e.g., barrier flanges) extending from the base panel **466** and between the first and second side panels **462**, **465**. As discussed above, the first and second side panels **462**, **465** are configured to enable coupling of the condensate capture receptacle **222** to the heat exchanger **112** (e.g., via the support brackets **226**). The shield panel **460**, the first and second side panels **462**, **465**, the base panel **466**, and the flange **468** may form a channel **474** (e.g., a basin, a cavity) within the condensate capture receptacle **222** to capture condensate blown off of the heat exchanger **112**, **300** by the air flow **114**, condensate falling, with respect to the vertical axis **104**, from the apex **212**, **304** of the heat exchanger **112**, **300**, condensate flowing along the shield panel **460**, the first side panel **462**, and/or the second side panel **465**, or any combination thereof. In other words, the condensate capture receptacle **222** forms the channel **474** and the flange **468** may retain (e.g., keep) the condensate captured within the channel **474**.

(78) Furthermore, the condensate captured may be directed towards the drain pan **220** via the one or more condensate ports **470** (e.g., discharge ports). In particular, the condensate may flow along the channel **474** and into the one or more condensate ports **470**, and then the condensate within the one or more condensate ports **470** may be directed by the force of gravity towards the drain pan **220**. In some embodiments, and as illustrated in FIG. **8**, the one or more condensate ports **470** may be disposed at one or more lateral ends **480** of the channel **474** and/or of the condensate capture receptacle **222**. Furthermore, the condensate collection assembly **122** may include a respective discharge conduit **476** fluidly coupled to each of the one or more condensate ports **470**. The discharge conduit **476** may extend from the respective condensate port **470** toward the basin **424** of the drain pan **220**. For example, each discharge conduit **476** may be a piece of hollow tubing (e.g., plastic tubing, rubber tubing) fitted over a respective open end **478** of the corresponding condensate port **470** to direct condensate from the channel **474** and into the basin **424** of the drain pan **220**. While one discharge conduit **476** is shown in FIG. **8**, it should be understood that each of the one or more condensate ports **470** may be equipped with a respective discharge conduit **476**. In some embodiments, the one or more condensate ports **470** may not be used to direct the condensate towards the drain pan **220**. For example, as discussed above, the heat exchanger **112** (e.g., the microchannel heat exchanger **300**) may be oriented in a vertical configuration such that condensate may flow downward, with respect to the vertical axis **104**, due to the force of gravity to the drain pan **220**. In such an implementation, the one or more condensate ports **470** may be capped (e.g., closed off), and the condensate capture receptacle **222** may be utilized to improve heat transfer efficiency of the heat exchanger **112** (e.g., the microchannel heat exchanger **300**), as discussed above.

(79) Furthermore, the condensate capture receptacle **222** may be formed from a single continuous piece of material, in some embodiments. For example, the condensate capture receptacle **222** may be formed from a plastic material and may be formed using an injection molding process. In particular, the use of a plastic to form the condensate capture receptacle **222** may improve production cost and, during operation, decrease corrosion of the condensate capture receptacle **222** that may be caused by exposure to condensate (e.g., water). Furthermore, a shape of the condensate

capture receptacle **222** may correspond to a geometry (e.g., a shape, a size) of the heat exchanger **112**. In particular, a geometry of the condensate capture receptacle **222** may be selected to generally correspond to a shape, geometry, or configuration of the apex **212** of the heat exchanger **112**. For example, and with reference to FIG. **6**, an angle of the head panel **464** (e.g., relative to the longitudinal axis **102**) may correspond to an angle of the first heat exchange portion **206**, while an angle of the base panel **466** (e.g., relative to the longitudinal axis **102**) may correspond to an angle of the second heat exchange portion **208**.

(80) Moreover, a width **472** of the shield panel **460** may correspond to a width of the apex **212**, **304** of the heat exchanger **112** (e.g., the microchannel heat exchanger **300**). In an installed configuration of the condensate capture receptacle **222** with the heat exchanger **112**, the shield panel **460** may be positioned at an offset distance **473** (e.g., a spaced arrangement) from the drain pan **220** along the vertical axis **104**. Thus, in the installed configuration, the shield panel **460** may be disposed within the air flow path **110** extending through the housing **108** of the HVAC unit **100**, as shown in FIGS. **5** and **6**. The shield panel **460** may extend across the air flow path **110** and/or across the apex **212**, **304** (e.g., bent portions **302** of the microchannel tubes **308**) along the lateral axis **106**. More specifically, the shield panel **460** may be positioned within the air flow path **110** downstream of the heat exchanger **112** (e.g., relative to the first direction **116** of the air flow **114**), such that the shield panel **460** overlaps with the apex **212**, **304** of the heat exchanger **112** (e.g., microchannel heat exchanger **300**) with respect to the longitudinal axis **102**. The corresponding shape of the condensate capture receptacle **222**, along with the adjustable positioning (e.g., mounting, securing) of the condensate capture receptacle **222** to the heat exchanger **112** so that the condensate capture receptacle **222** overlaps with the apex **212**, **304** of the heat exchanger **112** (e.g., microchannel heat exchanger **300**) and/or the bent portions **302** of the microchannel tubes **308**, results in efficient capture of condensate generated by the heat exchanger **112** (e.g., microchannel heat exchanger **300**) and improved heat exchange efficiency of the heat exchanger **112** (e.g., microchannel heat exchanger **300**). Additionally, the suspended position of the condensate capture receptacle **222** within the air flow path **110** enables reduced restriction of the air flow **114** (e.g., reduced drop in pressure due to blocking at least a portion of the air flow **114**) through the air flow path **110** of the HVAC unit **100**. Additional features and operation of the condensate capture receptacle **222** is described in further detail below with reference to FIGS. **9** and **10**.

(81) FIG. **9** is a perspective view of the embodiment of a portion of the HVAC unit **100**, illustrating a gap **500** (e.g., an opening) formed between the condensate capture receptacle **222** and the heat exchanger **112** in an installed configuration of the condensate collection assembly **122**. In particular, the gap **500** may be formed between the downstream end **210** of the second heat exchange portion **208** of the heat exchanger **112** and the flange **468**. During operation, the flange **468** may extend crosswise to the first direction **116** of air flow **114**. Furthermore, the gap **500** may enable condensate that may form on the second heat exchange portion **208** of the heat exchanger **112** (e.g., microchannel heat exchanger **300**) and that may travel along the second heat exchange portion **208** to the downstream end **210** of the second heat exchange portion **208** to enter the condensate capture receptacle **222**. For example, the air flow **114** blowing in the first direction **116** may cause condensate formed on the second heat exchange portion **208** of the heat exchanger **112** (e.g., microchannel heat exchanger **300**) to also move (e.g., travel, flow) in the first direction **116**, as indicated by arrow **502**, and towards the apex **212**, **304** and/or downstream end **210** of the heat exchanger **112** (e.g., microchannel heat exchanger **300**). The condensate may then enter into and be captured by the condensate capture receptacle **222**.

(82) It should be understood that the size of the gap **500** may be any suitable size so as to enable capture of condensate formed on an underside (e.g., with respect to the vertical axis **104**) of the heat exchanger **112** (e.g., the microchannel heat exchanger **300**) by the condensate capture receptacle **222**. In some embodiments, the size of the gap **500** may be selected by adjusting a mounted position (e.g., via the elongated slots **444** of the support brackets **226**) of the condensate

capture receptacle **222** with respect to the heat exchanger **112** (e.g., the microchannel heat exchanger **300**). To this end, the adjustable positioning of the condensate capture receptacle **222** provided by the elongated slots **444** of the support brackets **226** may enable the condensate capture receptacle **222** to maintain the gap **500** at a desired size while also enabling use of the condensate capture receptacle **222** with heat exchangers of varying sizes and/or tonnage. In addition, it should be understood that in some embodiments, an additional gap may be formed on a side of the condensate capture receptacle **222** opposite the gap **500**. For example, the additional gap may be formed between the first heat exchange portion **206** of the heat exchanger **112** (e.g., the microchannel heat exchanger **300**) and the condensate capture receptacle **222**. In some embodiments, the condensate capture receptacle **222** may be mounted to the heat exchanger **112** such that the gap **500** is not formed. Operation of the condensate capture receptacle **222** is described in further detail below with reference to FIG. **10**.

(83) FIG. **10** is a cross-sectional schematic side view of an embodiment of the heat exchanger **112** (e.g., the microchannel heat exchanger **300**) and the condensate capture receptacle **222**, illustrating capture of condensate **600** dislodged from the heat exchanger **112** (e.g., the microchannel heat exchanger **300**) via the air flow **114** traveling in the first direction **116**. Specifically, as the air flow **114** flows across and through the heat exchanger **112** (e.g., the microchannel heat exchanger **300**) in the first direction **116**, condensate **600** may be generated as moisture or water vapor within the air flow **114** condenses. In some instances, the condensate **600** may form and/or accumulate on the first and second heat exchange portions **206**, **208** (e.g., on the fins), or the first and second portions **310**, **314** of the microchannel tubes **308** (e.g., the fins **328**) of the microchannel heat exchanger **300**. As previously discussed and further illustrated in FIG. **10**, the air flow **114** may force the condensate **600** to travel generally in the first direction **116** and/or along the longitudinal axis **102**. For example, the condensate **600** may travel along the fins (e.g., the fins **328**) of the first and second heat exchange portions **206**, **208** (e.g., the first and second portions **310**, **314** of the microchannel tubes **308**). Some condensate **600** may ultimately reach one or more of the downstream ends **210** of the heat exchanger **112** (e.g., the microchannel heat exchanger **300**) and may be dislodged from the heat exchanger **112** (e.g., the microchannel heat exchanger **300**). The force of the air flow **114** may carry the condensate **600** from the downstream ends **210** of the heat exchanger **112** (e.g., the microchannel heat exchanger **300**). In some embodiments, the force of the air flow **114** may carry the condensate **600** from the bent portions **302** of the microchannel tubes **308** of the microchannel heat exchanger **300**. In any case, subsequently, the condensate **600** carried by the air flow **114** may contact the condensate capture receptacle **222** via the shield panel **460**, the first side panel **462**, the second side panel **465**, the head panel **464**, the base panel **466**, or any combination thereof. The captured condensate **600** may then be captured and collected within the channel **474** of the condensate capture receptacle **222**.

(84) As discussed above, the condensate capture receptacle **222** may also block and/or redirect one or more portions, as indicated by arrows **604**, **606**, of the air flow **114** traveling across the apex **212**, **304** of the heat exchanger **112** (e.g., the microchannel heat exchanger **300**). The redirected air flow portions indicated by arrows **604**, **606** may instead flow across the heat exchanger **112** at the first and/or second heat exchange portions **206**, **208** of the heat exchanger **112** (e.g., the first and/or second portions **310**, **314** of the microchannel heat exchanger **300**). Further, as discussed above, including the condensate capture receptacle **222** with the heat exchanger **112** (e.g., the microchannel heat exchanger **300**) may increase the overall heat transfer of the heat exchanger **112** by causing a greater percentage of the air flow **114** to travel across (e.g., interact with) finned portions of the heat exchange coils or tubes (e.g., the first and/or second heat exchange portions **206**, **208**, the first and/or second portions **310**, **314**) by blocking the air flow **114** from exiting the heat exchanger **112** via the fin-less portion of heat exchange coils or tubes (e.g., bent portions **302** at the apex **212**, **304**). Thus, the condensate capture receptacle **222** may improve the performance of the heat exchanger **112** (e.g., the microchannel heat exchanger **300**) during operation in addition

to efficiently and effectively capturing condensate generated by the heat exchanger **112**.

(85) In some embodiments, a portion or an entirety of the condensate capture receptacle **222** may be formed from a material that is generally or substantially impervious to water or liquid, but enables transmission of at least a portion of the air flow **114** therethrough. In another embodiment, a portion or an entirety of the condensate capture receptacle **222** may be formed from a material that blocks flow of both liquid and gas therethrough. In any case, the condensate **600** that impinges against the condensate capture receptacle **222** is diverted downwards, due to gravity, along a height **602** of the condensate capture receptacle **222** towards the channel **474**.

(86) FIGS. **11-14** illustrate view of an alternative embodiment of the condensate collection assembly **122**, in accordance with the presently disclosed techniques. In particular, the condensate capture receptacle **222** shown in FIGS. **11-13** and the support brackets **226** shown in FIG. **14** may provide additional adjustability to enable the condensate collection assembly **122** to be used in an installed configuration with heat exchangers **112** (e.g., microchannel heat exchangers **300**) that may vary in shape, geometry, size, capacity, or other aspects. For example, the condensate collection assembly **122** may be used with heat exchangers **112** that vary in a height **700** of the apex **304**, where the height **700** of the apex **304**, as shown in FIG. **6**, may be measured from a lower most point **702**, with respect to the vertical axis **104**, of the respective downstream end **210**, with respect to the first direction **116** of air flow **114**, of the second heat exchange portion **208** to an upper most point **704** of the respective downstream end **210** of the first heat exchange portion **206**.

(87) FIG. **11** is a perspective view of an embodiment of the condensate capture receptacle **222**, illustrating a multi-component configuration of the condensate capture receptacle **222**. In other words, the condensate capture receptacle **222** of the illustrated embodiment is not a single, integrally formed piece. In some embodiments, the condensate capture receptacle **222** may include a first translational piece **706** and a second translational piece **708**. The first and second translational pieces **706**, **708** may have complementary geometries such that the first translational piece **706** may translate (e.g., telescope, extend out, slide, move) along the second translational piece **708**, and/or vice versa, as indicated by arrows **710**. In particular, as illustrated, at least a portion of the second translational piece **708** may define a second partial perimeter **714** smaller than a first partial perimeter **716** of the first translational piece **706**. In this way, at least a portion of the second translational piece **708** may fit or extend within the first translational piece **706**. Thus, when the first and/or second translational pieces **706**, **708** are translated along or relative to one another, at least a portion of the second translational piece **708** is retained within the first translational piece **706**. While FIG. **11** shows the first translational piece **706** with the first partial perimeter **716** larger than the second partial perimeter **714** of the second translational piece **708**, and the first translational piece **706** enclosing at least a portion of the second translational piece **708**, it should be understood that in some embodiments, at least a portion of the first translational piece **706** may have a partial perimeter smaller than a partial perimeter of the second translational piece **708**. In such an embodiment, when the first and/or second translational pieces **706**, **708** are translated relative to one another, at least a portion of the first translational piece **706** may be retained within the second translational piece **708**.

(88) In addition, as discussed above, the condensate capture receptacle **222** may include the shield panel **460** (e.g., condensate barrier, capture panel, blocking plate), the first side panel **462** (e.g., on the first lateral side **450**), the second side panel **465** (e.g., on the second lateral side **452**), the head panel **464**, the base panel **466** opposite the head panel **464**, with respect to the vertical axis **104**, the one or more condensate ports **470** (e.g., drainage ports) fluidly coupled to the base panel **466** and formed therethrough, and the one or more flanges **468** (e.g., barrier flanges) extending from the base panel **466** and between the first and second side panels **462**, **465**. Furthermore, as shown in the illustrated embodiment, the first and second side panels **462**, **465** may include one or more elongated slots **712** formed therethrough. In particular, the one or more elongated slots **712** may enable adjustability of a mounting position of the condensate capture receptacle **222** relative to the

heat exchanger **112**.

(89) Furthermore, FIG. **12** is an exploded perspective view of the condensate capture receptacle **222** having a multi-component configuration. In particular, the first translational piece **706** may be formed of and include a first section **718** of the first side panel **462**, a first section **720** of the second side panel **465**, the head panel **464**, and a first section **722** of the shield panel **460**. Moreover, the second translational piece **708** may be formed of and include a second section **724** of the first side panel **462**, a second section **726** of the second side panel **465**, the base panel **466**, the one or more condensate ports **470** fluidly coupled to the base panel **466**, the one or more flanges **468** extending from the base panel **466** and between the second section **724** of the first side panel **462** and second section **726** of the second side panel **465**, and a second section **728** of the shield panel **460**. As illustrated, the first section **722** of the shield panel **460** includes one or more first holes **730** extending therethrough and arranged with a first set **732** of the one or more first holes **730** positioned within a first portion **736** of a total length **740** of the condensate capture receptacle **222**, and a second set **734** of the one or more first holes **730** positioned within a second portion **738** of the total length **740**. Specifically, in the illustrated embodiment, the first and second sets **732**, **734** of the one or more first holes **730** are aligned with one another relative to the lateral axis **106**. In addition, the second section **728** of the shield panel **460** includes one or more second holes **742** extending therethrough and arranged with a first set **746** of the one or more second holes **742** positioned within the first portion **736** of the total length **740** of the condensate capture receptacle **222**, and a second set **748** of the one or more second holes **742** positioned within the second portion **738** of the total length **740**. In particular, in the illustrated embodiment, the first and second sets **746**, **748** of the one or more second holes **742** are aligned with one another at an angle relative to the lateral axis **106**. In other words, each consecutive hole of the first set **746** of the one or more second holes **742**, starting from a first distal hole **750** and ending with a first proximal hole **752** with respect to a center **754** of the condensate capture receptacle **222**, decreases in a respective distance from a top edge **756** of the second section **728** of the shield panel **460**. In the same way, each consecutive hole of the second set **748** of the one or more second holes **742**, starting from a second distal hole **758** and ending with a second proximal hole **760** with respect to the center **754**, decreases in a respective distance from the top edge **756** of the second section **728** of the shield panel **460**.

(90) The angled arrangement of the first and second sets **746**, **748** of the one or more second holes **742** enables the first translational piece **706** to be secured (e.g., attached, fastened, connected) to the second translational piece **708** at multiple different extension (e.g., height, translational) arrangements (e.g., configurations), which enables formation of the condensate capture receptacle **222** in varying sizes (e.g., shapes). As discussed above, the size adjustability of the condensate capture receptacle **222** enables the condensate capture receptacle **222** to overlap with the apex **212**, **304** of different heat exchangers (e.g., microchannel heat exchangers) of varying sizes. For example, FIG. **13** illustrates side views of three extension arrangements of the first translational piece **706** in relation to the second translational piece **708** of the condensate capture receptacle **222**. In particular, a first extension arrangement **800** of the condensate capture receptacle **222** may include the first translational piece **706** and the second translational piece **708** positioned relative to one another such that the first and second distal most holes **750**, **758** of the second translational piece **708** align respectively with a first and second distal holes **810**, **812**, with respect to the center **754**, of the first and second sets **732**, **734** of the one or more first holes **730** of the first translational piece **706**. In the first extension arrangement **800**, the first and second translational pieces **706**, **708** may be fastened together by extending a fastener (e.g., a screw, a pin) through each respective alignment of the first distal hole **750** of the first set **746** of the one or more second holes **742** with the distal hole **810** of the first set **732** of the one or more first holes **730** and the second distal hole **758** of the second set **748** of the one or more second holes **742** with the second distal hole **812** of the second set **734** of the one or more first holes **730**.

(91) In addition, in the illustrated embodiment, the first translational piece **706** and the second translational piece **708** may be positioned relative to one another in a second extension arrangement **802** of the condensate capture receptacle **222**. The second extension arrangement **802** includes the first translational piece **706** and the second translational piece **708** arranged such that a first and second middle holes **806**, **808** of the second translational piece **708** align respectively with a first and second middle hole **814**, **816** of the first and second sets **732**, **734** of the one or more first holes **730** of the first translational piece **706**. In the second extension arrangement **802**, the first and second translational pieces **706**, **708** may be fastened together by extending a fastener (e.g., a screw, a pin) through each respective alignment of the first middle hole **806** of the first set **746** of the one or more second holes **742** with the first middle hole **814** of the first set **732** of the one or more first holes **730** and the second middle hole **808** of the second set **748** of the one or more second holes **742** with the second middle hole **816** of the second set **734** of the one or more first holes **730**.

(92) Furthermore, for the illustrated embodiment, a third extension arrangement **804** of the condensate capture receptacle **222** may include the first translational piece **706** and the second translational piece **708** positioned relative to one another such that the first and second proximal holes **752**, **760** of the second translational piece **708** align respectively with a first and second proximal hole **818**, **820**, with respect to the center **754**, of the first and second sets **732**, **734** of the one or more first holes **730** of the first translational piece **706**. In the third extension arrangement **804**, the first and second translational pieces **706**, **708** may be fastened together by extending a fastener (e.g., a screw, a pin) through each respective alignment of the first proximal hole **752** of the first set **746** of the one or more second holes **742** with the proximal hole **818** of the first set **732** of the one or more first holes **730** and the second proximal hole **760** of the second set **748** of the one or more second holes **742** with the second proximal hole **820** of the second set **734** of the one or more first holes **730**.

(93) It should be appreciated that while FIG. **13** illustrates three extension arrangements (e.g., the first, second, and third extension arrangements **800**, **802**, **804**) of the first and second translational pieces **706**, **708**, and further two sets (e.g., the first and second sets **732**, **734** of the one or more first holes **730**) of three holes extending through the first translational piece **706** and a corresponding two sets (e.g., the first and second sets **746**, **748** of the one or more second holes **742**) of three holes extending through the second translational piece **708**, the first and/or the second translational pieces **706**, **708** may include any number of sets (e.g., 1 set, 3 sets, 4 sets, etc.) of any number of holes (e.g., 1 hole, 2 holes, 4 holes, 5 holes, etc.) such that the first and the second translational pieces **706**, **708** may be positioned relative to one another in any number of extension arrangements (e.g., 2, 4, 5, 6, etc.).

(94) FIG. **14** is a perspective view of an embodiment of the support brackets **226** of the condensate capture receptacles **222**. In particular, as discussed above, the support brackets **226** may couple (e.g., directly couple) the condensate capture receptacle **222** to the heat exchanger **112** (e.g., the microchannel heat exchanger **300**) and position the condensate capture receptacle **222** such that the condensate capture receptacle **222** overlaps with (e.g., encloses, captures) the apex **212**, **304** of the heat exchanger **112** (e.g., the microchannel heat exchanger **300**). Furthermore, the support brackets **226** may support and suspend the condensate capture receptacle **222** above the drain pan **220** relative to the vertical axis **104**. In particular, with additional reference to FIG. **6**, the condensate capture receptacle **222** may extend from the first support bracket **230** to the second support bracket **232** and across the air flow path **110** of the air flow **114**. In addition, the support brackets **226** may couple the condensate capture receptacle **222** to the heat exchanger **112** (e.g., the microchannel heat exchanger **300**), despite a variation in size (e.g., shape) of the condensate capture receptacle **222** and/or the heat exchanger **112** (e.g., the microchannel heat exchanger **300**, the apex **212**, **304** of the heat exchanger/microchannel heat exchanger **112**, **300**). For example, and with additional reference to FIGS. **11** and **12**, the condensate capture receptacle **222** includes the one or more elongated slots

**712** extending through the first and second side panels **462**, **465**. As in the illustrated embodiment, the first section **718** and the second section **724** of the first side panel **462** may each include an elongated slot **712**, and the first section **720** and the second section **726** of the second side panel **465** may additionally each include an elongated slot **712**. In this way, the same support brackets **226** may couple (e.g., fastened) to the condensate capture receptacle **222** in a variety of extension arrangements (e.g., the first, second, and/or third extension arrangements **800**, **802**, **804**). Moreover, one or more fasteners **900** may extend through the alignment of one or more holes **902** extending through the support brackets **226** and the one or more elongated slots **712** of the condensate capture receptacle **222** to secure the support brackets **226** to the condensate capture receptacle **222**.

(95) As set forth above, embodiments of the present disclosure may provide one or more technical effects useful for efficiently capturing and/or collecting condensate that forms and/or accumulates on a heat exchanger and is dislodged from the heat exchanger by an air flow directed across the heat exchanger (e.g., in a generally horizontal direction). In particular, the condensate collection assembly discussed herein is configured to capture condensate, including condensate blowoff, that may be generated during operation of the heat exchanger, as well as provide improved heat exchange efficiency for the heat exchanger in an HVAC unit. It should be understood that the technical effects and technical problems in the specification are examples and are not limiting. Indeed, it should be noted that the embodiments described in the specification may have other technical effects and can solve other technical problems.

(96) While only certain features and embodiments have been illustrated and described, many modifications and changes may occur to those skilled in the art, such as variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, such as temperatures and pressures, mounting arrangements, use of materials, colors, orientations, and so forth, without materially departing from the novel teachings and advantages of the subject matter recited in the claims. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the disclosure.

(97) Furthermore, in an effort to provide a concise description of the exemplary embodiments, all features of an actual implementation may not have been described, such as those unrelated to the presently contemplated best mode, or those unrelated to enablement. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation specific decisions may be made. Such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure, without undue experimentation.

(98) The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . . ” or “step for [perform]ing [a function] . . . ”, it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).

## Claims

1. A condensate collection assembly, comprising: a drain pan configured to collect condensate generated by a heat exchanger of a heating, ventilation, and air conditioning (HVAC) system; and a condensate capture receptacle configured to discharge condensate to the drain pan, wherein the condensate capture receptacle comprises a first translational piece and a second translational piece,

- the first translational piece and the second translational piece are configured to overlap with the heat exchanger relative to a direction of air flow across the heat exchanger, are configured to couple to the heat exchanger, and are configured to be suspended above the drain pan relative to a direction of gravity, and the first translational piece is adjustable relative to the second translational piece.
2. The condensate collection assembly of claim 1, wherein the first translational piece and the second translational piece define a cavity configured to capture condensate dislodged from the heat exchanger by the air flow.
  3. The condensate collection assembly of claim 1, comprising a support bracket configured to couple the first translational piece and the second translational piece to the heat exchanger.
  4. The condensate collection assembly of claim 3, wherein the support bracket is a first support bracket, and the condensate collection assembly comprises a second support bracket configured to couple to the first translational piece and the second translational piece to the heat exchanger.
  5. The condensate collection assembly of claim 4, wherein the first translational piece and the second translational piece are configured to extend from the first support bracket to the second support bracket and across an air flow path of the air flow.
  6. The condensate collection assembly of claim 1, wherein the condensate capture receptacle is configured to couple to a downstream end of the heat exchanger, relative to the direction of air flow.
  7. The condensate collection assembly of claim 1, wherein the second translational piece comprises a barrier flange configured to extend along the heat exchanger in a direction crosswise to the direction of air flow such that an opening is formed between the barrier flange and the heat exchanger in an installed configuration of the condensate capture receptacle.
  8. The condensate collection assembly of claim 1, wherein the second translational piece defines a channel formed at a base of the second translational piece, wherein the channel is configured to direct condensate captured by the condensate capture receptacle toward the drain pan.
  9. The condensate collection assembly of claim 8, wherein the drain pan defines a basin configured to collect condensate generated by the heat exchanger, and the condensate capture receptacle comprises a discharge port configured to discharge condensate from the channel to the basin.
  10. The condensate collection assembly of claim 9, comprising a discharge conduit configured to be fluidly coupled to the discharge port, wherein the discharge conduit is configured to extend from the discharge port toward the basin and to direct condensate from the discharge port into the basin.
  11. The condensate collection assembly of claim 1, wherein the first translational piece is configured to couple to the second translational piece in a plurality of extension arrangements, and wherein a first size of the condensate capture receptacle in a first extension arrangement of the plurality of extension arrangements is smaller than a second size of the condensate capture receptacle in a second extension arrangement of the plurality of extension arrangements.
  12. A condensate collection system, comprising: a condensate trough configured to couple to a downstream end of a heat exchanger disposed within an air flow path of a heating, ventilation, and air conditioning (HVAC) system, relative to a direction of air flow across the heat exchanger, wherein the condensate trough is configured to capture condensate generated by the heat exchanger and is configured to be disposed offset from a drain pan of the HVAC system, the condensate trough comprises a first translational piece and a second translational piece, the first translational piece and the second translational piece are configured to overlap with the downstream end of the heat exchanger relative to the direction of air flow across the heat exchanger, are configured to couple to the heat exchanger, and are configured to be suspended above the drain pan relative to a direction of gravity, and the first translational piece is adjustable relative to the second translational piece.
  13. The condensate collection system of claim 12, comprising the drain pan, wherein the drain pan is configured to be positioned beneath the heat exchanger relative to the direction of gravity.

14. The condensate collection system of claim 13, wherein the drain pan comprises a base and a plurality of walls extending from the base to define an outer perimeter of the drain pan, wherein the condensate trough is configured to be disposed internal to the outer perimeter.
15. The condensate collection system of claim 12, comprising at least one bracket configured to couple to the heat exchanger and to the first translational piece and the second translational piece to secure the first translational piece and the second translational piece to the heat exchanger.
16. The condensate collection system of claim 12, wherein the second translational piece comprises a basin configured to capture condensate dislodged from the heat exchanger by an air flow and a discharge port configured to direct condensate from the basin toward the drain pan, and wherein the condensate collection system comprises a discharge conduit configured to fluidly couple to the discharge port and extend from the condensate trough toward the drain pan, wherein the discharge conduit is configured to direct condensate from the discharge port into the drain pan.
17. The condensate collection system of claim 12, wherein the first translational piece is configured to couple to the second translational piece in a plurality of translational arrangements, and wherein a first size of the condensate trough in a first translational arrangement of the plurality of translational arrangements is smaller than a second size of the condensate trough in a second translational arrangement of the plurality of translational arrangements.
18. A heating, ventilation, and air conditioning (HVAC) system, comprising: a heat exchanger disposed within an air flow path and configured to condition an air flow directed across the heat exchanger; and a condensate collection assembly configured to capture condensate generated by the heat exchanger and to discharge condensate to a drain pan disposed beneath the heat exchanger relative to a direction of gravity, wherein the condensate collection assembly comprises: a first translational piece; and a second translational piece coupled to the first translational piece, wherein the first translational piece and the second translational piece are configured to overlap with the heat exchanger relative to a direction of air flow across the heat exchanger along the air flow path, are configured to couple to the heat exchanger, and are configured to be suspended above the drain pan relative to the direction of gravity, and the first translational piece is adjustable relative to the second translational piece.
19. The HVAC system of claim 18, wherein the heat exchanger is a microchannel heat exchanger comprising microchannel tubes, a first portion of the microchannel tubes comprises a first set of fins, a second portion of the microchannel tubes comprises a second set of fins, the microchannel tubes comprise a bend between first portion and the second portion, and the first translational piece and the second translational piece are configured to overlap with the bend of the microchannel tubes relative to the direction of air flow across the heat exchanger.
20. The HVAC system of claim 18, comprising the drain pan, wherein the second translational piece defines a basin configured to capture condensate and a discharge port configured to direct condensate from the basin toward the drain pan, and wherein the condensate collection assembly comprises a discharge conduit configured to fluidly couple to the discharge port and extend from the second translational piece toward the drain pan at least partially along the direction of gravity, wherein the discharge conduit is configured to direct condensate from the discharge port into the drain pan.
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